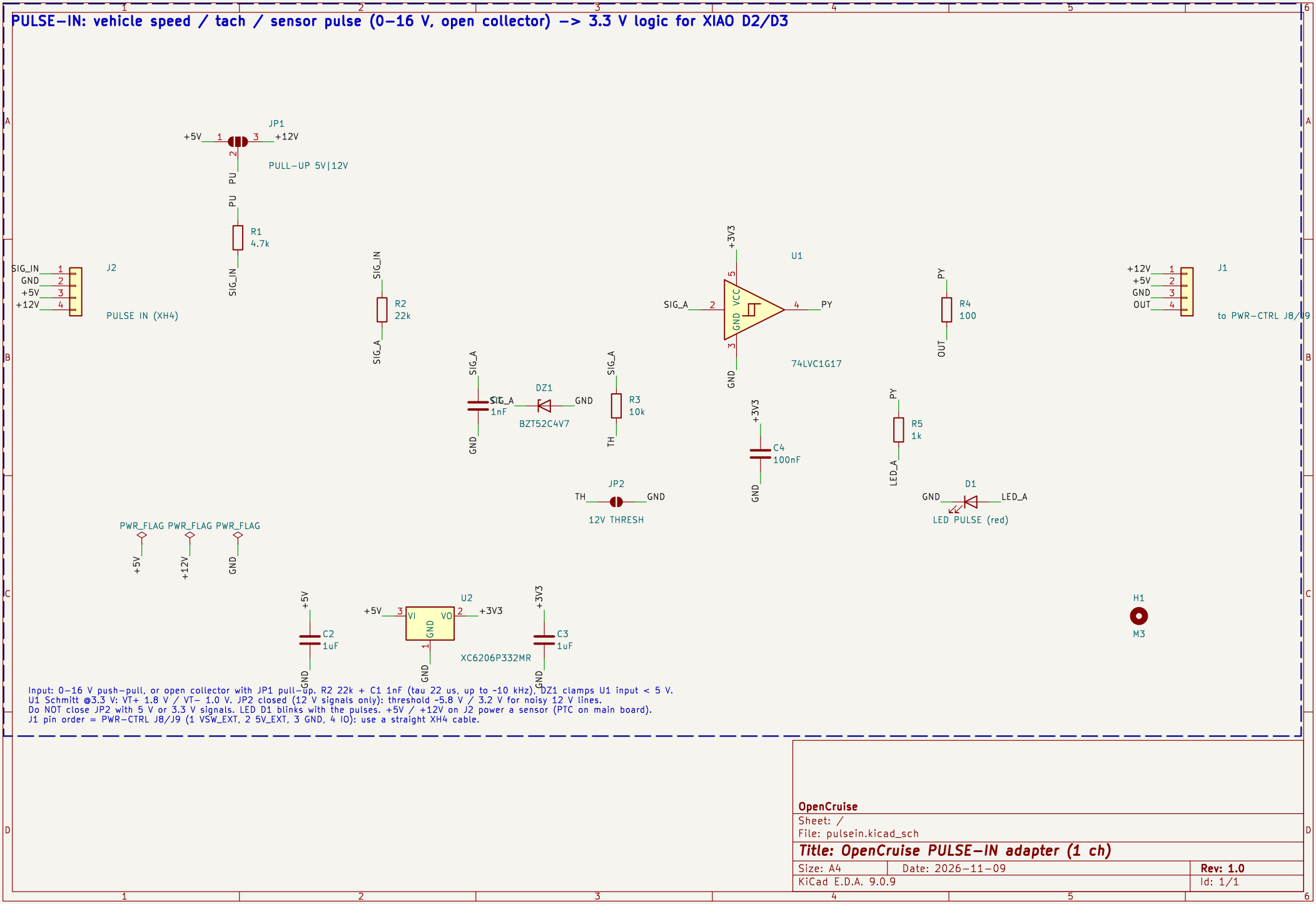

PULSE-IN: vehicle speed / tach / sensor pulse (0-16 V, open collector) -> 3.3 V logic for XIAO D2/D3
JP1
+5V
+12V
PULL-UP 5V|12V
PU
R1
4.7k
SIG_IN
J2
GND
PULSE IN (XH4)
R2
22k
SIG_A
U1
+3V3
VCC
74LVC1G17
PY
R4
100
OUT
J1
to PWR-CTRL J8/J9 (XH4)
DZ1
BZT52C4V7
1nF
R3
10k
TH
JP2
12V THRESH
C4
100nF
R5
1k
LED_A
D1
LED PULSE (red)
PWR_FLAG
U2
VI
VO
XC6206P332MR
C2
1uF
C3
H1
M3
Input: 0-16 V push-pull, or open collector with JP1 pull-up. R2 22k + C1 1nF (tau 22 us, up to ~10 kHz). DZ1 clamps U1 input < 5 V.
U1 Schmitt @3.3 V: VT+ 1.8 V / VT- 1.0 V. JP2 closed (12 V signals only): threshold ~5.8 V / 3.2 V for noisy 12 V lines.
Do NOT close JP2 with 5 V or 3.3 V signals. LED D1 blinks with the pulses. +5V / +12V on J2 power a sensor (PTC on main board).
J1 pin order = PWR-CTRL J8/J9 (1 VSW_EXT, 2 5V_EXT, 3 GND, 4 IO): use a straight XH4 cable.
OpenCruise
Sheet: /
File: pulsein.kicad_sch
Title: OpenCruise PULSE-IN adapter (1 ch)
Size: A4
Date: 2026-11-09
Rev: 1.0
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